

# Beethoven Clock — Electronics Assembly Guide

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# Purpose

This electronics assembly guide details descriptions, recommendations, and precautions for handling and assembling the electronic components of Beethoven Clock.

This Electronics Assembly Guide pertains to the following revisions of the PCBAs in Beethoven Clock:

- Beethoven Clock: Brain PCB – **Mk. II**
- Beethoven Clock: LED PCB – **Mk. I**
- Beethoven Clock: Flex PCB – **Mk. III**

The author of this document assumes no responsibility or liability for any errors, omissions, or outcomes resulting from assembling the electronic components of Beethoven Clock. Reader(s) are solely responsible for verifying the accuracy, applicability, and suitability of the information for their intended use. By using this document, the reader agrees to assume full responsibility for any risks associated with the application of the provided information.

# Full Electronics Assembly – Exploded View

# Full Electronics Assembly – Bill of Materials

| Drawing Item Number | Item Description                    | Item Manufacturer | Manufacturer Part Number | QTY |
|---------------------|-------------------------------------|-------------------|--------------------------|-----|
| 1                   | Beethoven Clock - Brain PCB, Mk. II | PCBWay            |                          | 1   |
| 2                   | Beethoven Clock - LED PCB, Mk. I    | PCBWay            |                          | 1   |
| 3                   | Beethoven Clock - Flex PCB, Mk. I   | PCBWay            |                          | 1   |
| 4                   | Brain-to-LED PCB Cable Assembly     | -                 | -                        | 1   |
| 5                   | LED-to-LED PCB Cable Assembly       | -                 | -                        | 1   |
|                     |                                     |                   |                          |     |

Brain-to-LED PCB Cable Assembly Part Number: **A08SUR08SUR32W254A**

LED-to-LED PCB Cable Assembly Part Number: **A08SUR08SUR32W51A**

^^If either of these are out of stock, try looking for the part number on [Octopart](#). Octopart compiles a bunch of part vendors. If not, you can use a different length, but make sure that the cables are the **reversed** version.

# Recommended Assembly Steps

## **1. Pre-Enclosure PCB test**

Be sure to connect and test all electronic components before assembling them inside of the clock. This includes:

- All **LEDs** on the LED PCB, including the AM/PM and alarm enable LEDs
- All **buttons** on the Brain PCB
- **Software functions** like time setting, alarm setting, and alarm sound
- All **cables** to ensure connectors don't have defects

**Caution: Never hot-plug anything.** That is to say – Connect power to the system **last**. Never plug/unplug any cables if the Brain PCB is powered.

**Caution:** When handling PCBs, try to touch the on-board components as little as possible, as this could transfer electrostatic discharge (ESD) into the components. The boards are designed to be able to handle ESD exposure, but extra caution can't hurt.

I assume you have the rest of the assembly steps solved. The same rules apply here:

- Make sure the Flex PCB makes solid contact with the screw terminals
- Apply hot glue to Flex PCB Screw Terminals
- Apply tape in between Flex PCB and Enclosure. Apply hot glue around Flex PCB perimeter to ensure it does not come loose (I've seen this happen on my end).
- If possible, hot glue the connectors of the Brain-to-LED and LED-to-LED cables once they are mated, but I imagine this may not be possible based on how the boards are inserted in the enclosure.